TerraMind on SmallMinesDS: Experiment Report

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Summary

We trained a shallow UNet decoder on SmallMinesDS 2022 (Sentinel-1 + Sentinel-2, 128 × 128 px) on top of TerraMind v1 tiny, either pretrained and frozen or randomly initialised and trainable, with and without D4 geometric augmentation, for N = 1 to 1,491 (all) training images (50 epochs, AdamW, cosine learning rate 5e-3 → 1e-5; 5 seeds for N ⩽ 250, 3 seeds for N ⩾ 500; 196 runs). The frozen pretrained encoder gives the higher test IoU at every N ⩾ 10 in every seed, reaching 0.745 with all images and augmentation against 0.673 for the encoder trained from scratch. With 10 images it already matches what the trainable encoder reaches with 250. Augmentation adds +0.02 to +0.04 IoU for the frozen encoder at every N ⩾ 3 and has no consistent effect on the trainable one, whose seed-to-seed spread is also 2–3× larger (Figure 2).

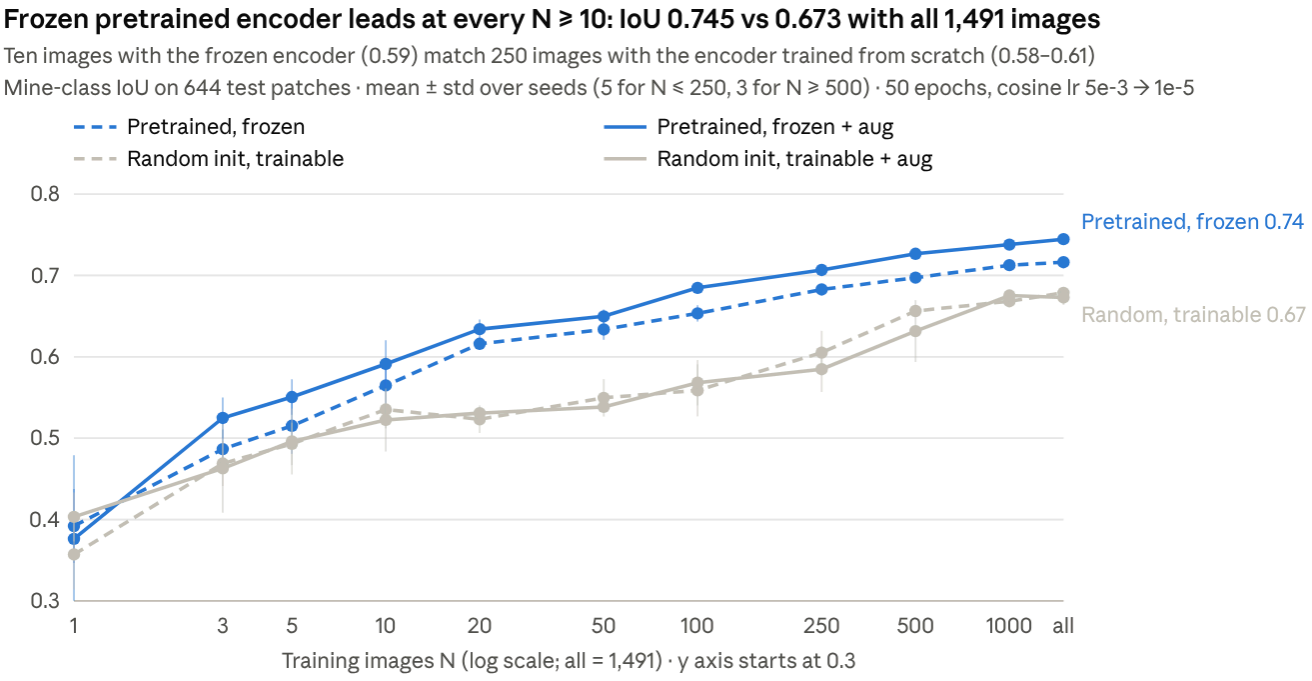


Figure 1. Mine-class IoU on the test set against the number of training images, mean ± std over seeds

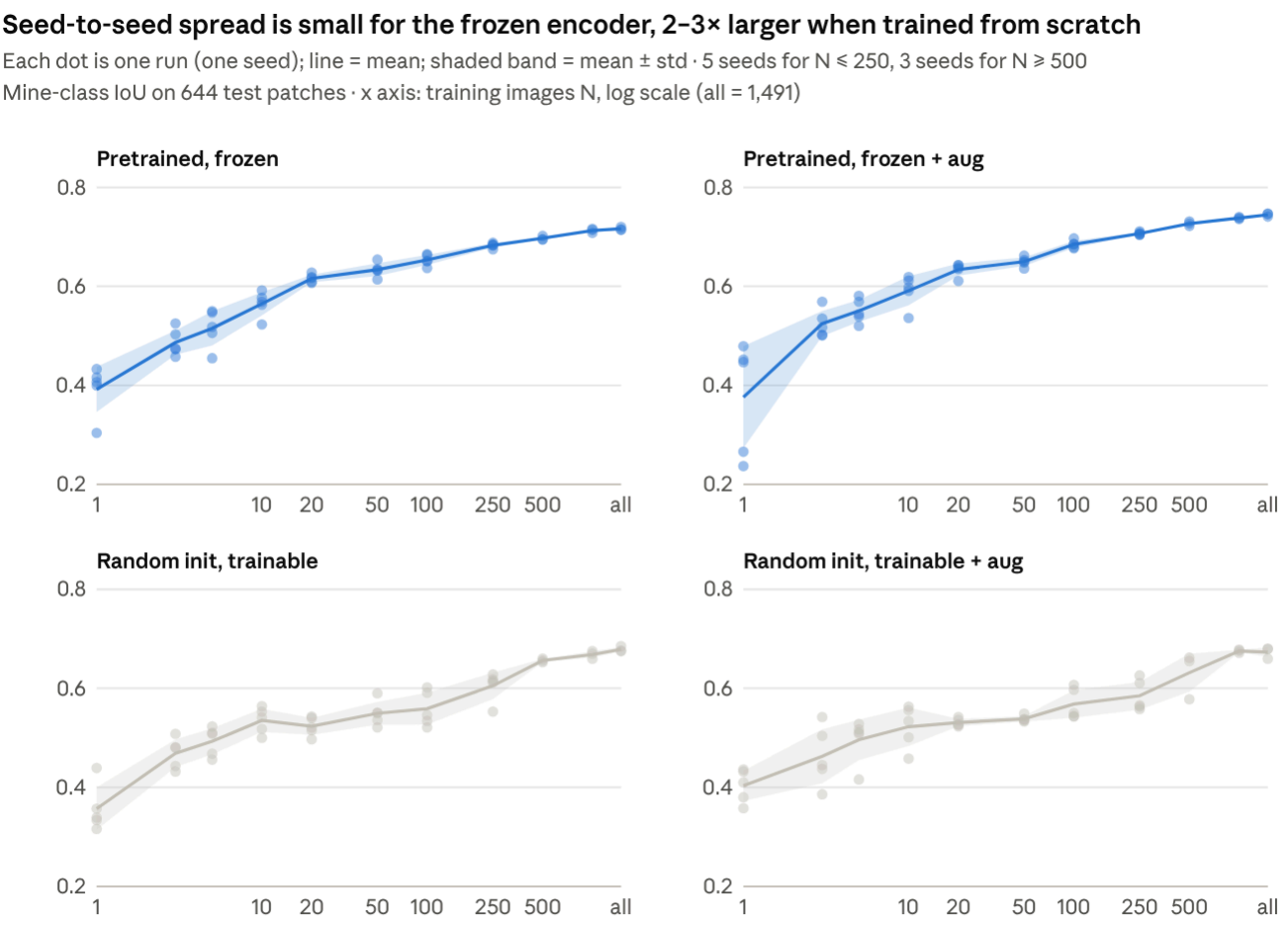


Figure 2. Seed-to-seed variance of IoU: every run as a dot, mean as a line, mean ± std as a shaded band

Table 1. Mine-class IoU on the test set (mean ± std over seeds; 5 seeds for N ⩽ 250, 3 seeds for N ⩾ 500).

N	Pretrained, frozen	Pretrained, frozen + aug	Random init, trainable	Random init, trainable + aug
1	0.392 ± 0.046	0.376 ± 0.103	0.357 ± 0.043	0.403 ± 0.030
3	0.487 ± 0.024	0.525 ± 0.025	0.469 ± 0.028	0.463 ± 0.054
5	0.515 ± 0.034	0.551 ± 0.022	0.493 ± 0.026	0.496 ± 0.040
10	0.565 ± 0.023	0.591 ± 0.029	0.535 ± 0.023	0.522 ± 0.039
20	0.616 ± 0.007	0.634 ± 0.012	0.523 ± 0.017	0.531 ± 0.007
50	0.633 ± 0.013	0.650 ± 0.008	0.550 ± 0.023	0.538 ± 0.005
100	0.653 ± 0.010	0.685 ± 0.007	0.559 ± 0.032	0.568 ± 0.028
250	0.683 ± 0.004	0.707 ± 0.002	0.605 ± 0.027	0.584 ± 0.028

N	Pretrained, frozen	Pretrained, frozen + aug	Random init, trainable	Random init, trainable + aug
500	0.697 ± 0.003	0.726 ± 0.003	0.656 ± 0.003	0.632 ± 0.038
1000	0.712 ± 0.003	0.738 ± 0.002	0.668 ± 0.006	0.675 ± 0.002
all (1,491)	0.716 ± 0.003	0.745 ± 0.003	0.679 ± 0.004	0.673 ± 0.009

Table 2. IoU gain from D4 augmentation (augmented minus unaugmented run with the same seed and training images; mean ± std over seeds, in brackets the seeds that improved).

N	Pretrained, frozen	Random init, trainable
1	−0.016 ± 0.073 (3/5)	+0.046 ± 0.044 (4/5)
3	+0.038 ± 0.016 (5/5)	−0.006 ± 0.045 (2/5)
5	+0.035 ± 0.027 (5/5)	+0.003 ± 0.034 (3/5)
10	+0.027 ± 0.010 (5/5)	−0.013 ± 0.024 (2/5)
20	+0.018 ± 0.007 (5/5)	+0.008 ± 0.017 (3/5)
50	+0.016 ± 0.005 (5/5)	−0.011 ± 0.024 (2/5)
100	+0.031 ± 0.007 (5/5)	+0.010 ± 0.007 (5/5)
250	+0.024 ± 0.003 (5/5)	−0.021 ± 0.034 (2/5)
500	+0.029 ± 0.002 (3/3)	−0.025 ± 0.036 (1/3)
1000	+0.025 ± 0.002 (3/3)	+0.007 ± 0.008 (2/3)
all (1,491)	+0.029 ± 0.005 (3/3)	−0.006 ± 0.014 (2/3)

Code, per-run results and full details: github.com/asimarslan11/SmallMinesDS-Experiment-Project.